Universidade do Minho · Escola de Engenharia · Mestrado em Bioinformática

Metabolic Modelling of

Aspergillus niger & Grapevine

Pathogenic Interactions

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Context - The Threat to Viticulture

Vitis vinifera

- Portugal's cultural & economic heritage;
- Berry undergoes véraison — sugar accumulation, acid decrease, cell wall softening;
- Innate defences: stilbenes, phytoalexins (resveratrol, ε-viniferins).

Aspergillus niger

- Opportunistic phytopathogen — infects from véraison through post-harvest;
- Cell wall-degrading enzymes + organic acid secretion (citrate, gluconate);
- Mycotoxins: Ochratoxin A (OTA), fumonisins.

Chemical fungicides dominate management
environmental impact drives urgent need for sustainable alternatives

Motivation - What Is Missing?

GEMs Exist in Isolation

- iJB1325 (A. niger) and iMS7199 (V. vinifera) were developed independently.
- No compartmentalised host-pathogen community model existed.

No Dynamic Simulation

- The temporal evolution of infection — biomass accumulation, sucrose depletion, leakage dynamics — was never modelled.

Virulence and Defense

- Mycotoxin biosynthesis costs and the mechanistic link between véraison-associated disruption and susceptibility during infection remained unquantified.

Objectives

01. Curation & Validation

Targeted curation of iJB1325 (A. niger) and iMS7199 (V. vinifera, mature + green). Stoichiometric consistency verified via Memote

02. Nutrient Cross-feeding

Compartmentalised community model (T1/T2 transport). Quantify the Metabolic Interaction Potential (MIP) between host and pathogen.

03. Fungal Vulnerabilities

Single-gene and single-reaction knockout under infection nutritional constraints. Identify essential metabolic hubs as biocontrol targets

Extensions

- dFBA dynamic infection simulation
- Green vs mature berry dFBA comparison
- 2D parametric sensitivity — 49-run grid scan
- OTA/kotanin growth–virulence tradeoff analysis
- Green→Mature→Post-harvest full trajectory
- Custom metabolic interface map (matplotlib)

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Materials & Methods - Computational Framework

01. Model Curation

- iJB1325 (A. niger)
- iMS7199 (V. vinifera)
  - mature berry
  - green berry

02. Memote Validation

- Stoichiometric consistency

03. Community Model

- Infection compartment
- T1/T2 transport reactions

04. dFBA Simulation

- SOA
- 432h trajectory

05. Vulnerability Analysis

- KO
  - Genes
  - Reactions
- Mycotoxins Interface map

Tools: COBRApy · Gurobi (academic) · Memote · Matplotlib · Python 3.10

Models: iJB1325 (Brandl et al., 2018) · iMS7199 (Sampaio et al., 2024)

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Objective 01.

Curation and Validation of Metabolic Models

Memote

iJB1325 — A. niger
total score 29.1%
passed: 31
failed: 27
passed: 1

iMS7199 — V. vinifera mature & green
total score 4.0%
passed: 30
failed: 14
passed: 15

iJB1325 — A. niger

- ATP transport reaction r2137 (ATP <=> ATPp) was identified as permitting thermodynamically implausible peroxisomal ATP backflow and curated by constraining its lower bound from −1000 to 0, enforcing unidirectional cytosol-to-peroxisome transport.
- Curation without affect growth rate.

iMS7199 — V. vinifera mature & green

In both models:

- 218 reaction IDs were prefixed with R_ to resolve Gurobi LP variable name conflicts,
- Compartment codes C001–C010 were remapped to semantic identifiers and missing compartment names added where absent in the original models
- Curation without affect bounds or growth rate.

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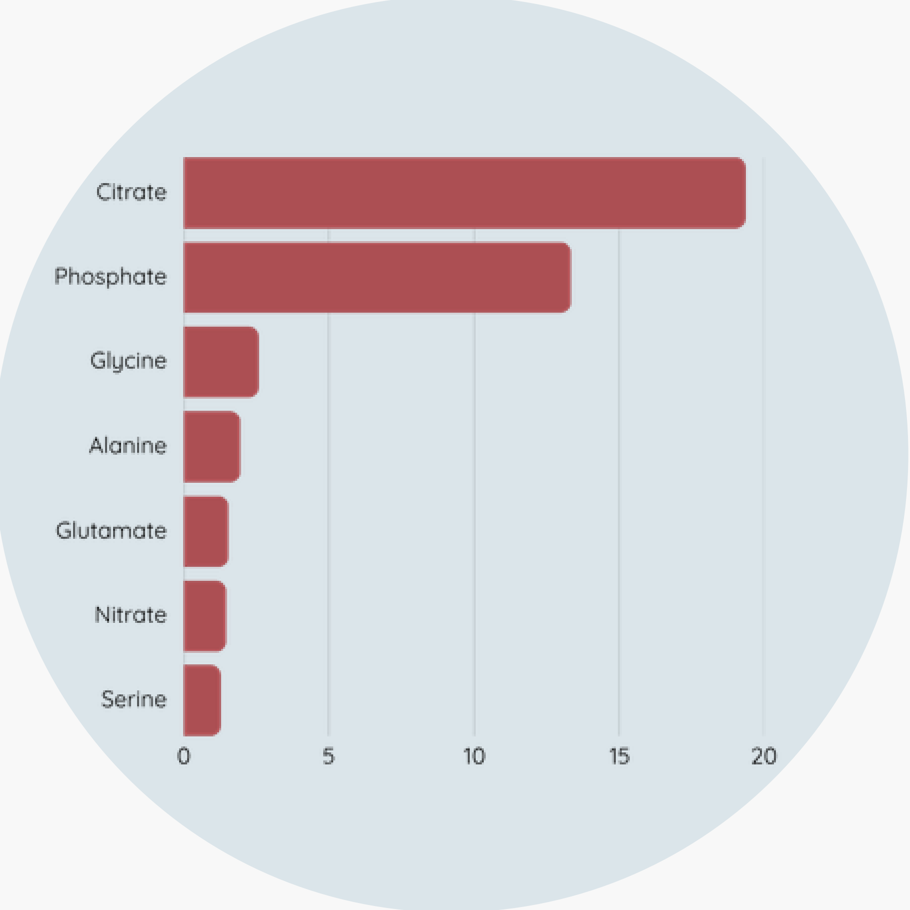
Metabolic Interation Potencial



T1 proxy flux — mature berry, f_infection = 0.95
Total: 42.1 mmol/gDW/h
Citrate dominates (46%)

Objective 02.

Compartmentalised Community Model & MIP

V. vinifera (cytosol_plant) —**T1**→ **Infection Compartment** —**T2**→ **A. niger (extracellular)**

- **prefix_model():** vv_ + an_ prefix all reactions and metabolites
- **INTERFACE_EXACT:** 19 metabolites defined
- **T1:** UB = proxy × leakage × f_infection
- **T2:** UB = proxy × leakage × f_infection × mm_suc
- **R_BOUNDARY_an_* blocked:** uptake only via T2
- **community.optimize():** 2 decoupled FBAs per timestep (SOA)

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proxy = maximum metabolite flux at optimal FBA (iMS7199)
leakage = fraction of host metabolite flux release into infection compartment
f_infection = infection severity factor (0,1)
mm_suc = Michaelis-Menten half-saturation constant for sucrose uptake

Objective 03.

Metabolic Vulnerabilities & Biocontrol Targets

Growth–Virulence Tradeoff

OTA (Ochratoxin A)
36.5% growth loss

Kotanin
47.4% growth loss

Each +0.1 mmol/gDW/h of forced secretion:

Knockout Methodology

- Infection medium: T2 bounds with proxy × leakage × f_infection
- find_essential_genes(community, threshold=0.05) — COBRApy
- find_essential_reactions(community, threshold=0.05) — COBRApy
- Criterion: growth < 5% of wild-type after KO
- Screened for host homology: targets lacking vine orthologs prioritized
- Green vs mature comparison: essentiality gain at véraison

Essential Metabolic Hubs — Biocontrol Targets

- **Glycolysis (EMP pathway)**
  - Essential under all infection conditions
- **TCA cycle reactions**
  - Critical for citrate secretion + biomass
- **Phosphate (uptake)**
  - +0.556 h^{-1} — most impactful single nutrient (comparable to all other infection nutrients combined)

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Results - Véraison as the Critical Infection Window

54h

t_f95 — mature berry
(time to 95% infection)

70h

t_f95 — green berry
(16h longer)

0.556 h^{-1}

Growth increase
from phosphate alone

47.1 g/L

Peak A. niger biomass
(post-harvest, Day 6)

Metabolic Signature Shift at Véraison

Dominant T1 metabolite switches from glutamate (green, 18.15 mmol/gDW/h) to citrate (mature, 19.37 mmol/gDW/h). TCA-centred metabolism post-véraison favours A. niger.

Defence Pathway Collapse

ANS/LDOX gene gains essentiality exclusively in the mature model. Stilbene biosynthesis (RXN-87) carries zero flux — resveratrol defence effectively blocked.

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Results - Dynamic Infection Simulation & Robustness

3-Phase dFBA Trajectory (432 h)

Green (0–48h)

- 20% leakage · $\delta = 0.003\ h^{-1}$
- Slow growth under stilbene pressure

Mature (48–144h)

- 100% leakage · $\delta = 0.002\ h^{-1}$
- Rapid exponential growth

Post-harvest (144–432h)

- Sucrose depletion ($t_{1/2}$ ≈ Day 9)
- Biomass collapse: 47.1 → ~0 g/L

Sensitivity Analysis (3C) — 49-run grid

Primary metric: t_f95
(time to f_infection = 0.95)

- half_sat = 0.001 — t_f95 = 18h
- half_sat = 0.010 (baseline) — t_f95 = 54 h
- half_sat = 0.025 — t_f95 = 102 h
- half_sat ≥ 0.050 — NaN

Key finding: t_f95 is K-independent by construction (threshold = 19 × half_sat).
Qualitative conclusions robust across all 49 parameter combinations.

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Limitations & Future Work

Limitations

- half_sat, leakage factors and K_fungus are phenomenological — not experimentally validated
- No host-pathogen homology screening performed — target specificity unconfirmed (nomenclature incompatibility)
- All results are computational hypotheses pending experimental validation

Future Work

- **Host homology screening** (priority): resolve ID incompatibility (NCBI numeric vs BioCyc) and perform BLAST of essential fungus GPR genes vs. grapevine proteome to confirm biocontrol target specificity
- **In vivo validation:** quantify fungal biomass, calibrate model parameters, and test knockdown of essential targets post-homology screening

Conclusions

Véraison is the critical window

- Mature berry reaches near-complete infection 16h faster (54h vs 70h).
- Metabolic shift from glutamate (green) to citrate (mature) as dominant T1 flux - Total T1 proxy: 42.1 mmol/gDW/h

Defence suppression explained

- ANS/LDOX - essentiality only in mature model.
- RXN-87 (resveratrol) with zero stilbene flux in mature model provides a GEM-based mechanistic link between véraison and A. niger susceptibility

Phosphate is the dominant growth driver

- +0.556 h^{-1} from phosphate alone — comparable to all other infection nutrients combined.
- Key target for biocontrol intervention.

Infection is self-limiting post-harvest

- Sucrose exhaustion drives biomass collapse (day 18).
- Sucrose: $t_{1/2} \approx$ Day 9; peak biomass 47.1 g/L

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